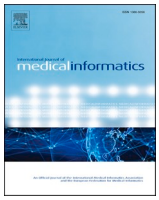
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Extending CARDIO:DE: Additional annotation guidelines and evaluation of NLP approaches for clinical applications

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ABSTRACT

Background: Cardiovascular diseases are a major cause of morbidity and mortality, and the management of these conditions generates extensive clinical data. The CARDIO:DE dataset, a German-language corpus of cardiovascular clinical routine letters, has been developed to support natural language processing research. This study seeks to enhance the dataset by introducing refined annotation guidelines and expanding the annotation schema.
Objective: The objective of this study was to extend the CARDIO:DE dataset with additional annotation categories, and evaluate state-of-the-art NLP models to enhance the utility of the dataset for clinical applications.
Methods: The annotation schema was expanded to include categories such as diagnostic procedures, medical finding, and therapeutic interventions (Diagnostic, Diagnosis, Drug, Medical_Finding, Therapy). The iterative annotation process involved expert annotators, ensuring high-quality, consistent annotations. Four models—GBERT, medBERT.de, XLM-RoBERTa, and TinyLlama—were fine-tuned and evaluated on the dataset. Model performance was assessed using entity-wise precision, recall, and F1 scores.
Results: The extended dataset includes 304,582 token-based annotations, with the highest concentration in medical finding. The inter-annotator agreement scores improved during the iterative process, reaching up to 0.98 for certain subsets. Among the evaluated models, TinyLlama outperformed the other models in entity recognition, achieving a macro-average F1 score of 0.845, highlighting its potential for clinical NLP tasks.
Conclusions: The extended CARDIO:DE dataset, with its refined annotation guidelines provides a robust foundation for natural language processing applications in the clinical domain. The performance of the TinyLlama model demonstrates the potential of fine-tuning non-domain-specific models for clinical text processing. This work paves the way for more accurate NLP solutions in healthcare, particularly for information extraction and decision support in cardiology.

1. Introduction

Cardiovascular diseases (CVDs) are the leading cause of morbidity and mortality globally, accounting for approximately 4.1 million deaths within European Society of Cardiology member countries alone [3]. The clinical management of CVDs generates vast amounts of unstructured data, such as patient anamneses, diagnoses, and treatment plans, typically stored in free-text format within electronic health records [2].

The initial development of the CARDIO:DE dataset aimed to address the need for a structured annotated corpus to facilitate natural language processing (NLP) research in the German-speaking cardiovascular domain. Current annotation layers predominantly target the extraction of drug-related data, such as dosage and strength. However, this study advances the dataset by expanding annotation categories to include additional medical entities. These entities can be codified in downstream tasks using classifications such as ICD (International Classification of Diseases) enabling a broader and more detailed representation of each patient's entire treatment regimen [18–22].

In the field of German-language clinical NLP, one significant limitation is the scarcity of annotated medical texts, particularly in specialized areas like cardiovascular care. Consequently, much of the research to date has relied on general-purpose BERT models that lack

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domain-specific supervised finetuning for clinical narratives. Addressing these challenges, the objective of our research is to develop an annotated corpus, that can act as a gold standard for finetuning German models for the clinical domain.

2. Methods

2.1. Dataset description CARDIO:DE

CARDIO:DE is the first freely available German clinical corpus dedicated to the cardiovascular domain, developed to support research in natural language processing. This unique dataset contains 500 clinical routine doctor's letters from the Heidelberg University Hospital, collected between 2020 and 2021 [1]. The corpus is divided into two subsets: CARDIO:DE400, with 400 documents and full annotations, and CARDIO:DE100, with 100 documents retained as a held-out test set for future shared tasks. This split allows for both training and evaluation of machine learning models, fostering reproducible research in the field.

Accessible via the Heidelberg University heiDATA repository [dataset] [11].

2.2. Annotation workflow

To enhance the CARDIO:DE dataset, a structured iterative annotation process was implemented to optimize the accuracy and reliability of the annotations [4] (Appendix A). This process was conducted using the INCEpTION [5] annotation tool, which provided comprehensive oversight and enabled effective quality control. Initially, about 15 % of the corpus was annotated by two experts in their respective fields—one a pharmacist with extensive clinical experience, the other a health economist. The purpose of this phase was to identify discrepancies and establish a consistent baseline for annotation [12]. To evaluate the agreement between the annotators, the Krippendorff's Alpha unitizing score was used [13]. This statistical measure is particularly suited to data that involves variable annotation borders. A threshold of 0.85 was set as the desired level of agreement, reflecting the high accuracy needed for clinical annotations, where even minor errors could have significant consequences for downstream NLP applications [6]. After this initial phase, the annotators undertook a cross-review process, during which they carefully discussed and resolved the discrepancies that had been identified. These discussions were thoroughly documented, to guide further annotation. The overall annotation process is illustrated in Fig. 1: *Annotation Workflow*, which provides a visual representation of the key steps, including initial annotation, discrepancy resolution, and refinement. Following these revisions, the agreement score was recalculated to determine whether the improvements were effective.

To further validate the robustness of the process, an additional 5 % of the corpus was annotated with overlap. As the agreement score declined by just 0.01, remaining below our predefined threshold of 0.05, we deemed the overlapping annotation phase complete. At this point, the remaining corpus was annotated independently by each annotator. The entire annotation process took seven months to complete. The time required to annotate each document ranged from 10 to 40 min, depending on the complexity of the text and the nature of the required annotations.

2.3. Expanded annotation schema

Building on the initial guidelines, a more comprehensive annotation schema was developed to capture the nuances of clinical narratives more effectively. The labels correspond to standard medical coding systems used in the German healthcare system: ICD for diagnoses, OPS for diagnostic and therapeutic procedures, and ATC for pharmaceuticals. Additionally, 'Medical Findings' could later be mapped to the corresponding SNOMED CT codes. The extended schema included the following labels, which were largely inspired by other corpora [14–17]:

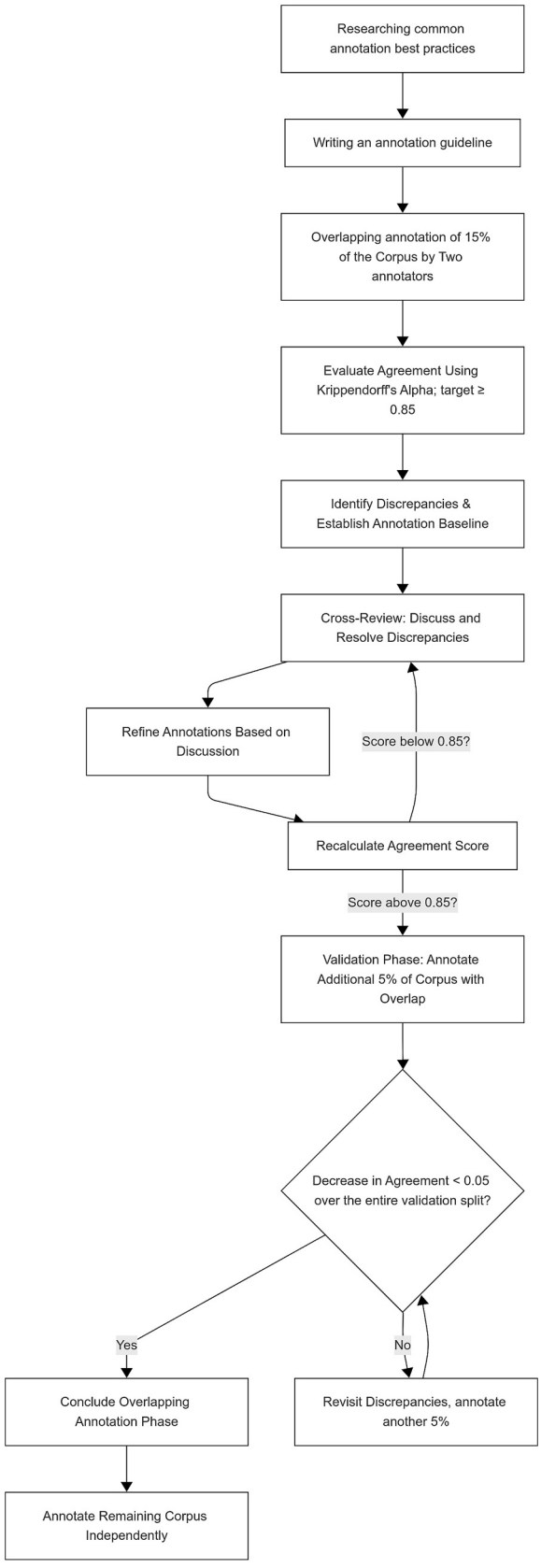


Fig. 1. Annotation workflow.

Diagnosis: This label was assigned to all entities that could be encoded with an ICD code, even if the entity did not fully represent a fine grained ICD Code. For example, a diagnosis such as "Type 1 Diabetes Mellitus" would be labeled as "Diagnosis" even though the entity does not provide any information about organ complications or whether it is controlled.

Diagnostic: This label was applied to procedures and actions aimed at diagnosing conditions, including medical histories, laboratory tests, imaging studies, and diagnostic surgical interventions. The *Diagnostic* label was also used for general statements implying diagnostic procedures, such as "cardiological follow-up examination" (Fig. 2).

Drug: The *Drug* label was used to annotate the specific name of a pharmaceutical substance, whether identified by its international nonproprietary name or its brand name. This label included all substances with pharmacological, metabolic, or immunological effects.

Medical Finding: This label was assigned to objective data obtained through diagnostics, such as laboratory results, imaging findings, electrocardiograms, and clinical scores. The *Medical Finding* label focused on what was specifically identified, as opposed to the *Diagnostic* label, which described the method of identification (Fig. 3).

Therapy: This label covered interventions primarily intended for the treatment, alleviation, or prevention of diseases, including the administration of specific medications as part of a therapeutic regimen.

2.4. Guidelines for special cases

To enhance the precision of annotation, guidelines were established for handling special cases, including (Justifications can be found in the annotation guidelines; see Appendix A):

Negations: Negations are not annotated unless they appear in the middle of the span, where omitting them would impair understanding. For example, "While lying down, no pericardial effusion" would include the negation as part of the "Medical Finding" annotation.

Speculative Statements: Speculative statements, such as those indicating uncertainty or future possibilities, were generally not annotated unless they provided critical context.

Lateralities and Qualifiers: Annotations were refined to include lateralities (e.g., "left," "right") and qualifiers (e.g., "severe," "moderate") to capture the full clinical picture.

Overlapping Entities: In cases where overlapping entities occurred, only the entity with the higher informational content was annotated to avoid redundancy and maintain clarity [14].

3. Results

3.1. Quantitative assessment

The annotation process resulted in substantial improvements in the quality and consistency of the annotations. Across the corpus, a total of 304,582 token-based annotations were made, with *Medical Finding* comprising approximately two-thirds of these annotations Table 1.

The inter-annotator agreement (IAA) scores improved significantly through the iterative process (Appendix A):

Table 2 shows the IAA scores for both the CARDIO:400 and the CARDIO:100 datasets through the iterative annotation process as described in section 1.2. While not directly comparable with scores determined via F1 score calculation for IAA, these scores indicate a high level of agreement among annotators, validating the effectiveness of the extended annotation guidelines and the iterative annotation process.

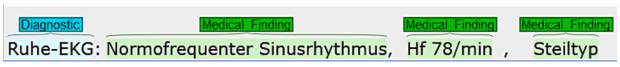


Fig. 3. Example of the medical finding annotation label.

3.2. Qualitative assessment

The qualitative analysis of the annotations provided several key insights: One of the main findings was the presence of minor discrepancies in label assignments. These discrepancies often occurred with phrases that could be interpreted in different ways depending on the context. For instance, the phrase "80 % main stem stenosis" could be annotated as either a *Diagnosis* or a *Medical Finding*, depending on the emphasis placed within the clinical context. In instances where spans of two entities with distinct classes overlapped, annotators prioritized labeling the class deemed to hold greater informational value. Specifically, when a medical finding overlapped with a diagnosis, the diagnosis was selected. Similarly, in cases where a therapeutic intervention overlapped with a diagnostic procedure, the therapy was chosen. When both a drug and a therapy overlapped, the therapy was consistently prioritized. Another common source of variation was the determination of the exact length of the spans to be annotated. Discrepancies were frequently observed when deciding whether to annotate entire phrases or just their individual components. For example, one annotator marked the entire phrase "detection of E. Coli," while the other annotated just "E. Coli." These differences were addressed through in-depth discussions and the creation of detailed documentation to guide the annotation process more uniformly.

The analysis of average label lengths highlights the complexity associated with different annotation categories. Longer labels, particularly within *Medical Findings*, tend to result in higher disagreement among annotators due to the increased likelihood of variations in identifying entity boundaries [14]. For instance, consider the sentence: "*Tiefe Unterschenkelvenen soweit beurteilbar komprimierbar oder mit antegrad provozierbarem Flussnachweis, ohne Hinweise für intraluminales thrombotisches Material*" which exemplifies a complex Medical Finding span where differences in boundary selection can lead to annotation inconsistencies. This complexity can negatively impact model performance, as any mislabeling within a long entity causes it to be evaluated as entirely incorrect. Despite these challenges, *Medical Findings* achieved higher entity-specific F1 scores during model inference compared to *Therapy* and *Diagnosis*, likely due to consistent labeling of repetitive and detailed findings during cross-review. In contrast, shorter labels in categories such as *Therapy* and *Diagnosis* may reduce annotation complexity but exhibit lower F1 scores, possibly reflecting differences in repetition frequency and contextual clarity.

3.3. Benchmark

As an initial benchmark, four distinct transformer models were fine-tuned and evaluated on the task of extracting medical data from the dataset (Appendix B, Appendix C). The first model, GBERT (deepset/gbert-base) [7], is a German BERT model, previously used by Richter-Pechanski et al. [1] in their original publication of the CARDIO:DE dataset. The second model, medBERT.de (GerMedBERT/medbert-512) [8], also BERT-based, has been trained specifically on German medical texts, including clinical notes, research papers, and healthcare-related documents. The third model, XLM-RoBERTa (FacebookAI/xlm-roberta-large) [9], is an evolved version of the BERT architecture, optimized with a larger and more diverse training corpus. Although XLM-RoBERTa is not specialized in the medical domain, its broad linguistic capabilities

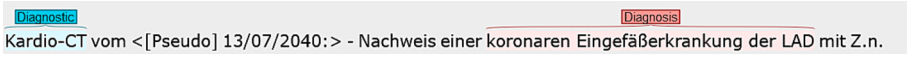


Fig. 2. Example of the diagnosis and diagnostic annotation labels.

Table 1
Word and entity count for each label.

Label	CARDIO:DE			CARDIO:DE 400			CARDIO:DE 100		
	Words	Entities	Word density	Words	Entities	Word density	Words	Entities	Word Density
Total	1,124,833			912,582			212,251		
No label	820,251			668,869			151,382		
Total with labels	304,582	90,534	27,1 %	243,713	73,546	26,7 %	60,869	16,988	28,7 %
Diagnosis	31,807	13,948	2,8 %	24,704	11,152	2,7 %	7,103	2,796	3,3 %
Diagnostic	21,226	11,327	1,9 %	16,963	9,171	1,9 %	4,263	2,156	2,0 %
Drug	7,271	6,839	0,7 %	5,772	5,389	0,63 %	1,499	1,450	0,71 %
Medical-Finding	201,026	46,976	17,9%	161,835	38,605	17,7%	39,191	8,371	18,5%
Therapy	43,252	11,444	3,8 %	34,439	9,229	3,8 %	8,813	2,215	4,2 %

Table 2
IAA scores for CARDIO:DE 400 after 65 and 83 overlapping annotations respectively, CARDIO:DE 100 after 15 and 20 overlapping annotations respectively.

IAA after 65 letters CARDIO:DE 400			**IAA after 15 letters CARDIO:DE 100**		
	1st Iteration	2nd Iteration		1st Iteration	2nd Iteration
IAA Score	0,79	0,90	IAA Score	0,91	0,98
IAA after 83 letters CARDIO:DE 400			**IAA after 20 letters CARDIO:DE 100**		
	1st Iteration			1st Iteration	
IAA Score	0,89		IAA Score	0,97	

made it a compelling choice for comparison. Lastly, TinyLlama (TinyLlama/TinyLlama-1.1B-intermediate-step-1431 k-3 T), a compact, open-source model derived from the Llama2 family, was included. Unlike the other models, TinyLlama is a decoder-based autoregressive model, primarily designed for text generation tasks and was trained on a mix of natural language and code data, with no specific focus on medical contexts [10].

TinyLlama presented a unique challenge, as it lacked built-in capabilities for named entity recognition (NER). To address this, a custom method was developed to generate training data for fine-tuning. Our approach used a template format, encoding each word with its corresponding label in the form of 'word\tlabel\n'. The model was trained to learn this pattern and accurately predict the correct labels. An example data point is shown below:

'### Context\nKeine relevante Stenosewirkung, geringe Insuffizienz nachweisbar.\n\n.

Answer\n\nKeine\tO\nrelevante\tB-MEDICAL_FINDING\nStenosewirkung\tI-MEDICAL_FINDING\n,\tO\ngeringe\tB-MEDICAL_FINDING\nInsuffizienz\tI-MEDICAL_FINDING\nnachweisbar\tO\n.\tO\n\n</s>'.

This process allowed us to explore the flexibility and adaptability of TinyLlama within the medical domain, despite its initial limitations.

Here '### Answer' is the response term that indicates the expected model answer, when receiving the first part of this prompt. This is used with a DataCollatorForCompletionOnlyLM to train the model just on the expected answers. The model answer is then scanned for the learned pattern and the words and labels are extracted. This creates a new kind of error for this model type, when the words of the prompt are not correctly replicated. However, it has shown, that the replication of the correct words, labels and patterns is learned sufficiently well within the first epoch. The evaluation of the given answers of the different models is done by using the evaluate python library in conjunction with the seqeval method [23] with the standard IOB2 scheme. This enables entity-wise comparison putting high emphasis on the correct recognition of entity boundaries. For model training we used an 80:20 split for the size of the training and validation data. The models were evaluated each epoch during fine-tuning and the best epoch, with the highest macro average F1, was tested against the CARDIO:DE 100 test dataset.

Table 3 shows that the TinyLlama model outperforms the other three models across all class types by an average of about 9,4 percent points in micro average F1 and about 8,5 percent points in macro average F1 scores. With its 1.1B parameters it is about ten times bigger than the two BERT models and about double the size of the large RoBERTa model, so the increase in performance should not surprise. However, this shows the usefulness of text generation models for the NER task. Between the other three models the performance is quite similar. The basic GBERT model is building a solid baseline, with medbert.de and XLMRoberta being on par with each other. It seems that the smaller medBERT.de model can compensate for the lack in size with domain specific knowledge.

3.4. Error analysis

Fig. 4 provides a detailed breakdown of error types by model, revealing distinct patterns between TinyLlama and the BERT-based models. While all models show a predominance of errors related to incorrect span identification, TinyLlama exhibits significantly fewer false positives, wrong label assignments and overall span boundary mistakes. This suggests that TinyLlama is less prone to hallucination in entity detection and better at consistently labelling semantically matching entities. The BERT models, conversely, display a quite similar distribution of error types.

The confusion matrices in Fig. 5 further illustrate label-specific challenges shared across the BERT models. All models commonly confuse MEDICAL_FINDING and DIAGNOSIS, reflecting the inherent difficulty in clearly differentiating these categories despite careful efforts to distinguish the two labels during the annotation process. Though significantly smaller, the second most common error is confusion between the labels THERAPY, DIAGNOSIS and MEDICAL_FINDING, revealing overlapping semantic boundaries in the annotation schema.

In summary, the error analysis underscores the strengths and weaknesses of generative and classification-based models for clinical entity recognition. TinyLlama's fewer false positives, boundary mistakes and label errors, despite illegal label generation, suggest a robust underlying capability for detection of entity boundaries and types.

4. Discussion

The extension of the CARDIO:DE dataset represents a significant advancement in clinical natural language processing in the German language domain, addressing some of the limitations of its original iteration. By expanding the annotation schema and introducing additional guidelines, the enhanced dataset offers a more comprehensive representation of clinical language in cardiology. This enables researchers to develop more robust NLP models capable of handling the intricacies of clinical text with greater accuracy.

One of the key achievements of this project are the high IAA scores, which demonstrate the reliability of the annotations. Achieving such high scores, particularly with complex clinical data, underscores the effectiveness of the iterative annotation process and the importance of expert involvement in annotation tasks. The inclusion of annotators with

Table 3
. Ntity-wise precision (Pr), recall (Re) and F1-score (F1) for medical entity recognition across five distinct class types utilizing four different models.

	GBERT			medBERT.de			XLMRoberta			TinyLlama		
Class type	Pr	Re	F1	Pr	Re	F1	Pr	Re	F1	Pr	Re	F1
DIAGNOSIS	0.719	0.689	0.704	0.743	0.711	0.727	0.744	0.720	0.732	0.786	0.785	**0.785**
DIAGNOSTIC	0.757	0.787	0.772	0.771	0.792	0.781	0.778	0.785	0.782	0.859	0.841	**0.850**
DRUG	0.878	0.928	0.902	0.902	0.941	0.921	0.901	0.939	0.920	0.983	0.951	**0.967**
MEDICAL_FINDING	0.685	0.777	0.728	0.696	0.780	0.735	0.699	0.775	0.735	0.821	0.859	**0.840**
THERAPY	0.610	0.672	0.640	0.641	0.689	0.664	0.641	0.678	0.659	0.765	0.805	**0.785**
Micro Average	0.705	0.763	0.733	0.722	0.772	0.746	0.725	0.769	0.746	0.827	0.845	**0.836**
Macro Average			0.749			0.766			0.765			**0.845**

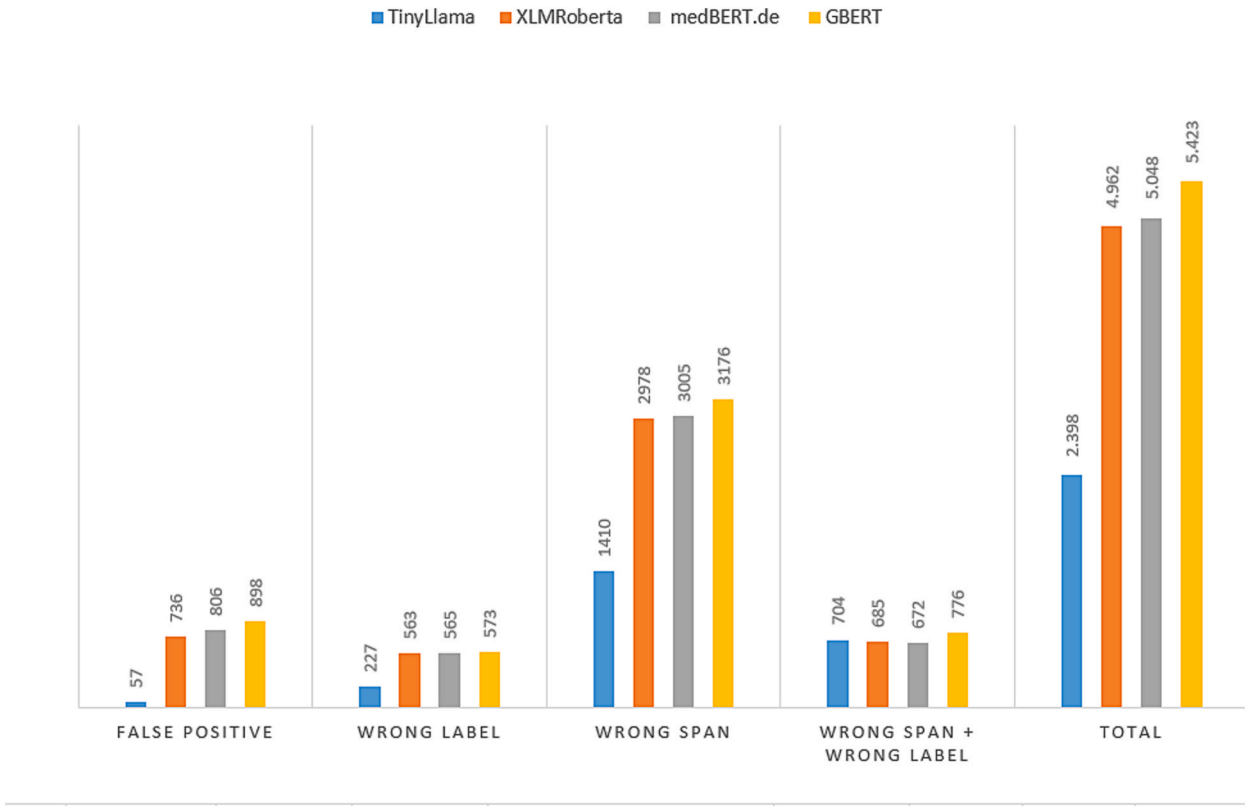


Fig. 4. Number of errors per type and model.

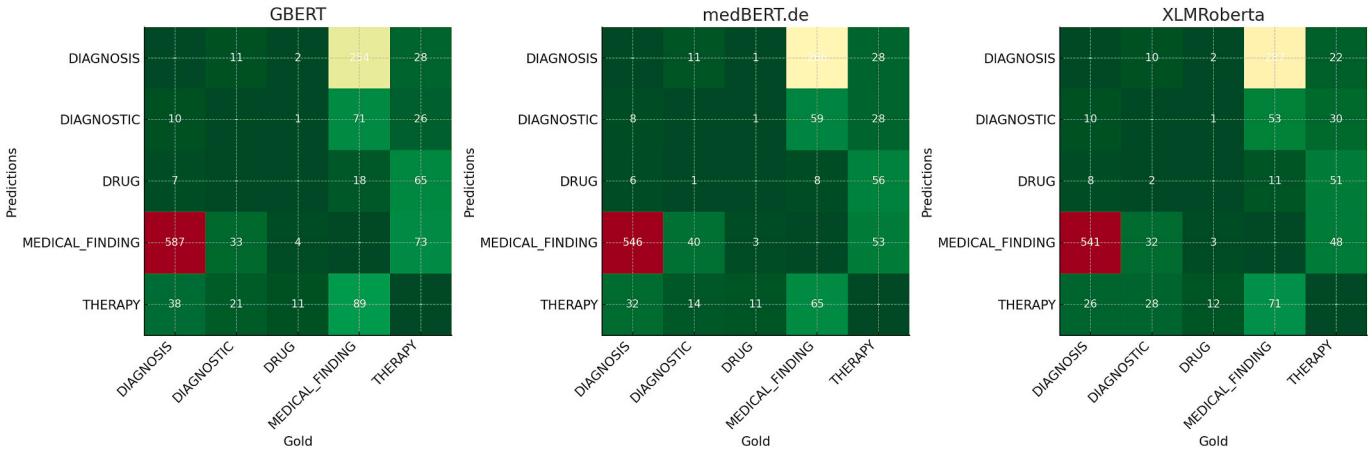


Fig. 5. Confusion matrix.

healthcare backgrounds ensured that discrepancies were resolved efficiently, leading to consistent and accurate annotations across the dataset. Furthermore, the expanded annotation schema enables more fine-grained differentiation between diagnosis and medical findings, as well as between diagnostic and therapeutic procedures. This enhances the dataset's applicability across various NLP tasks, including Named Entity Recognition (NER), information extraction for automated coding and billing support through ICD code assignment, automated selection of clinical trial cohorts based on inclusion and exclusion criteria, and clinical decision support tasks such as guideline adherence verification.

The model benchmarking conducted as part of this study demonstrates the dataset's potential for fine-tuning and evaluating transformer-based models. The performance of the TinyLlama model, which outperformed other models despite its lack of built-in medical context training, highlights the potential of non-specialized text generation models in the clinical domain when appropriately fine-tuned. This finding opens the door to further exploration of open-source models for clinical NLP tasks, which may offer flexible and scalable solutions for medical text processing. Nevertheless, some challenges remain. The handling of ambiguous phrases and span length discrepancies still requires further refinement. Although the current guidelines have significantly reduced these issues, continued work is needed to ensure consistency, particularly as the dataset is applied to more complex tasks or different clinical contexts. Moreover, the complexity of medical narratives means that future expansions of the dataset may need to incorporate additional annotation layers or explore domain-specific language models for enhanced performance.

There is also potential of bigger open source LLM's such as Llama 70B/405B or similar models for automatic or semi-automatic annotation workflows. While this study focused on creating a new human-annotated gold standard for the German medical domain, we plan to investigate the annotation consistency and scalability of synthetic and semi-synthetic annotation methods using various language models and NLP pipelines compared against our curated dataset. This future work aims to assess whether these methods can accelerate annotation processes without sacrificing quality, especially in regards to very complex entities. A detailed analysis of energy consumption during training and inference is provided in Appendix D.

5. Conclusions

This study presents an alternative approach to annotating the CARDIO:DE dataset by introducing refined annotation guidelines and an expanded schema, enhancing its utility for clinical NLP applications. The iterative annotation process and expert involvement ensured high-quality annotations with strong inter-annotator agreement, enabling a robust foundation for clinical NLP tasks. Benchmarking with diverse NLP models demonstrated the potential of modern transformer-based approaches, with TinyLlama achieving the highest performance. These results highlight the importance of combining expert-driven annotation processes with state-of-the-art machine learning techniques to unlock the potential of unstructured medical data. The extended CARDIO:DE dataset offers a valuable resource for future research and development in the German clinical domain, fostering advancements in healthcare data analysis and decision support.

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CRediT authorship contribution statement

Matthias Becker: Writing – original draft, Supervision, Project administration, Conceptualization. **Mario Krumscheid:** Writing – review & editing, Validation, Methodology, Investigation, Data curation. **Alisa Knobelspies:** Writing – review & editing, Data curation. **Markus Seydel:** Writing – review & editing, Formal analysis, Data curation. **Phillip Richter-Pechanski:** Writing – review & editing. **Alexander Karl:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Summary Table

- Additional Annotation Schema for Clinical NLP: The study extends the CARDIO:DE dataset by introducing refined annotation guidelines and an expanded schema, capturing nuanced clinical details across categories such as diagnostics, medical findings, therapies, and drugs, with high inter-annotator agreement scores (>0.9).
- Evaluation of Diverse NLP Models: Four state-of-the-art NLP models, including GBERT, medBERT.de, XLM-RoBERTa, and TinyLlama, were fine-tuned and benchmarked on the dataset. TinyLlama demonstrated the best performance, achieving a macro-average F1 score of 0.845, highlighting the potential of advanced transformer models in clinical applications.
- Advancing Clinical Data Processing: This work provides a robust resource and methodological benchmark for processing German clinical narratives, paving the way for more accurate and effective NLP solutions in healthcare to improve information extraction, clinical decision support, and quality analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2025.106009.

References

[1] P. Richter-Pechanski, P. Wiesenbach, D.M. Schwab, et al., A distributable German clinical corpus containing cardiovascular clinical routine doctor's letters, Sci. Data 10 (2023) 207, https://doi.org/10.1038/s41597-023-02128-9.
[2] T.A. Koleck, et al., Natural language processing of symptoms documented in free-text narratives of electronic health records: a systematic review, J. Am. Med. Inform. Assoc. 26 (4) (2019) 364–379.
[3] A. Timmis, et al., European society of cardiology: cardiovascular disease statistics 2021, Eur. Heart J. 43 (2022) 716–799.
[4] J. Pustejovsky, A. Stubbs, Natural language annotation for machine learning: a guide to corpus-building for applications, O'Reilly Media, Inc.", ", 2012.
[5] J.-C. Klie, M. Bugert, B. Boullosa, R. Eckart de Castilho, I. Gurevych, The INCEpTION platform: Machine-assisted and knowledge-oriented interactive annotation. in proceedings of system demonstrations of the 27th international conference on computational linguistics (COLING Santa Fe, New Mexico, USA, 2018.
[6] A. Sylolypavan, et al., The impact of inconsistent human annotations on AI driven clinical decision making, npj Digital Med. 6 (1) (2023) 26.
[7] B. Chan, S. Schweter, T. Möller, German's next Language Model. 10.48550/arXiv (2020) 2010.10906.
[8] Bressem, Keno & Papaioannou, Jens-Michalis & Grundmann, Paul & Borchert, Florian & Adams, Lisa & Liu, Leonhard & Busch, Felix & Xu, Lina & Loyen, Jan &

Niehues, Stefan & Augustin, Moritz & Grosser, Lennart & Makowski, Marcus & Aerts, Hugo & Löser, Alexander. (2023). MEDBERT.de: A Comprehensive German BERT Model for the Medical Domain. 10.48550/arXiv.2303.08179.

[9] Conneau, Alexis & Khandelwal, Kartikay & Goyal, Naman & Chaudhary, Vishrav & Wenzek, Guillaume & Guzman, Francisco & Grave, Edouard & Ott, Myle & Zettlemoyer, Luke & Stoyanov, Veselin. (2020). Unsupervised Cross-lingual Representation Learning at Scale. 8440-8451. 10.18653/v1/2020.acl-main.747.

[10] Zhang, Peiyuan, et al. "Tinyllama: An open-source small language model." arXiv preprint arXiv:2401.02385 (2024).

[11] Christoph Dieterich, 2022, "CARDIO:DE [V1.01]", https://doi.org/10.11588/data/AFYQDY, heiDATA, V7.

[12] Roberts A, Gaizauskas R, Hepple M, Davis N, Demetriou G, Guo Y, Kola J, Roberts I, Setzer A, Tapuria A, Wheeldin B. The CLEF corpus: semantic annotation of clinical text. AMIA Annu Symp Proc. 2007 Oct 11;2007:625-9. PMID: 18693911; PMCID: PMC2655900.

[13] KRIPPENDORFF, Klaus. Computing Krippendorff`s alpha-reliability [online]. 2011.

[14] Florian Borchert, Christina Lohr, Luise Modersohn, Jonas Witt, Thomas Langer, Markus Follmann, Matthias Gietzelt, Bert Arnrich, Udo Hahn, and Matthieu-P. Schapranow. 2022. GGPONC 2.0 - The German Clinical Guideline Corpus for Oncology: Curation Workflow, Annotation Policy, Baseline NER Taggers. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3650–3660, Marseille, France. European Language Resources Association.".

[15] Johnson, A., Pollard, T., & Mark, R. (2016). MIMIC-III Clinical Database (version 1.4). PhysioNet. https://doi.org/10.13026/C2XW26.

[16] Madeleine Kittner, Mario Lamping, Damian T Rieke, Julian Götze, Bariya Bajwa, Ivan Jelas, Gina Rüter, Hanjo Hautow, Mario Sänger, Maryam Habibi, Marit Zettwitz, Till de Bortoli, Leonie Ostermann, Jurica Ševa, Johannes Starlinger, Oliver Kohlbacher, Nisar P Malek, Ulrich Keilholz, Ulf Leser (2021). Annotation and initial evaluation of a large annotated German oncological corpus JAMIA Open, Volume 4, Issue 2, ooab025.

[17] L. Modersohn, S. Schulz, C. Lohr, U. Hahn, GRASCCO - the first publicly shareable, multiply-alienated German clinical text corpus, Stud. Health Technol. Inform. 17 (296) (2022 Aug) 66–72, https://doi.org/10.3233/SHTI220805. PMID: 36073490.

[18] Dai H, Wang C, Chen C, Liou C, Lu A, Lai C, Shain B, Ke C, Wang W, Mir T, Simanjuntak M, Kao H, Tsai M, Tseng V Evaluating a Natural Language Processing–Driven, AI-Assisted International Classification of Diseases, 10th Revision, Clinical Modification, Coding System for Diagnosis Related Groups in a Real Hospital Environment: Algorithm Development and Validation Study J Med Internet Res 2024;26:e58278 URL: https://www.jmir.org/2024/1/e58278 DOI: 10.2196/58278.

[19] L. Liu, et al., Automated ICD coding using extreme multi-label long text transformer-based models, Artif. Intell. Med. 144 (2023) 102662.

[20] Kaur, Rajvir, Jeewani Anupama Ginige, and Oliver Obst. "A systematic literature review of automated ICD coding and classification systems using discharge summaries." arXiv preprint arXiv:2107.10652 (2021).

[21] Simmons A, Takkavatakarn K, McDougal M, Dilcher B, Pincavitch J, Meadows L, Kauffman J, Klang E, Wig R, Smith G, Soroush A, Freeman R, Apakama DJ, Charney AW, Kohli-Seth R, Nadkarni GN, Sakhuja A. Benchmarking Large Language Models for Extraction of International Classification of Diseases Codes from Clinical Documentation. medRxiv [Preprint]. 2024 Nov 23: 2024.04.29.24306573. doi: 10.1101/2024.04.29.24306573. Update in: Appl Clin Inform. 2024 Nov 28. doi: 10.1055/a-2491-3872. PMID: 39606395; PMCID: PMC11601733.

[22] Boyle, Joseph S., et al. "Automated clinical coding using off-the-shelf large language models." arXiv preprint arXiv:2310.06552 (2023).

[23] Nakayama, H. (2018). seqeval: A testing framework for sequence labeling (Version 1.2.2) [Computer software]. GitHub. Retrieved from https://github.com/chakki-works/seqeva.